

PRACTICAL-04

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>


int main()
{
    int i;
    pid_t pid;


    printf("Parent process started. PID = %d\n", getpid());


    // Create 3 child processes
    for (i = 1; i <= 3; i++)
    {
        pid = fork();

        if (pid < 0)
        {
            perror("fork failed");
            exit(1);
        }

        if (pid == 0)
        {
            printf("Child %d: PID = %d, Parent PID = %d\n",
                i, getpid(), getppid());

            sleep(i);
```

```

        printf("Child %d finished.\n", i);

        exit(0);
    }
}

// Parent waits for all children

for (i = 1; i <= 3; i++)
{
    wait(NULL);

    printf("Parent collected a child process.\n");
}

printf("All child processes completed.\n");

return 0;
}

```

OUTPUT:

```

sohan_jadala@DESKTOP-OI7PGP2:~$
sohan_jadala@DESKTOP-OI7PGP2:~$ ls
2520080060_OSSP  Fork  OSLab  OSSP_Lab  Practical  Skill
sohan_jadala@DESKTOP-OI7PGP2:~$ cd 2520080060_OSSP
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP$ ls
2520080060_ForgeOS  2520080060_Practical  2520080060_Skill  OS_C0-1_Practice_Programmes.pdf
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP$ cd 2520080060_Practical
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ ls
Practical-01  Practical3  Practical3.c
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ gedit Practical4.c

** (gedit:628): WARNING **: 08:54:45.141: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:628): WARNING **: 08:54:45.141: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ gcc Practical4.c -o Practical4
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ ./Practical4
Parent process started. PID = 1004
Child 1: PID = 1005, Parent PID = 1004
Child 2: PID = 1006, Parent PID = 1004
Child 3: PID = 1007, Parent PID = 1004
Child 1 finished.
Parent collected a child process.
Child 2 finished.
Parent collected a child process.
Child 3 finished.
Parent collected a child process.
All child processes completed.

```